

a4a final runs - draft

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Data input for the assessment

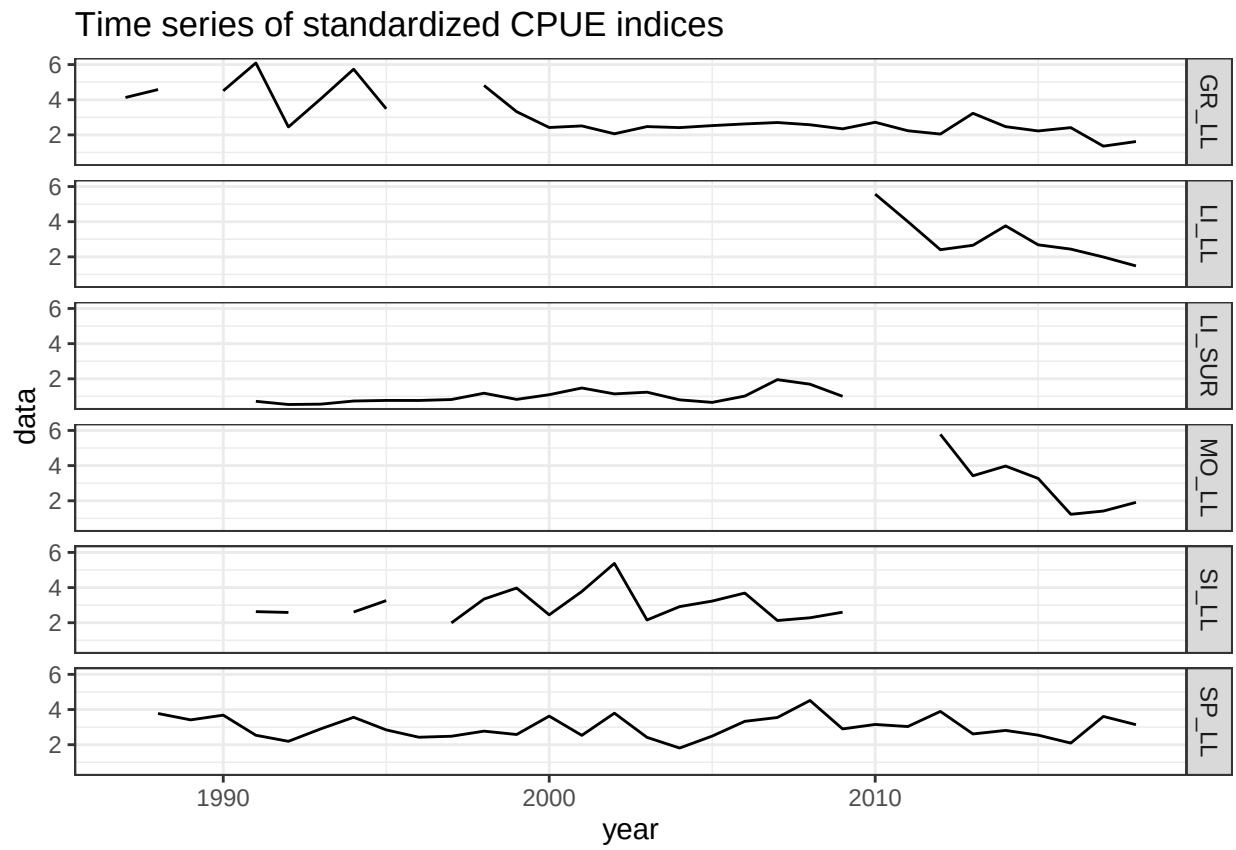
Catch Data

- Catch-at-age data since 1985
- plus group 5
- $\bar{f}$ at 2 - 4

CPUE Data

- Six standardized CPUE series
 - Greek longliners since 1987
 - Moroccan longliners since 2012
 - Spanish longliners since 1988
 - Ligurian longliners since 2010
 - Sicilian longliners from 1991 to 2009
 - Ligurian surface longliners 1991 to 2009
- Representative of age groups 2 - 4

CPUE indices



Model setup

- Fishing mortality $\mathbf{F}$

$$F : \sim s(year, k = 17) + s(age, k = 3)$$

- Recruitment $\mathbf{R}$

$$R : \sim s(year, k = 15)$$

- catchability $\mathbf{Q}$

$$Q : \begin{cases} \sim 1, & \text{Greek longline} \\ \sim 1, & \text{Spanish longline} \\ \sim s(year, k = 3), & \text{Moroccan longline} \\ \sim s(year, k = 3), & \text{Ligurian longline} \\ \sim 1, & \text{Sicilian longline} \\ \sim s(year, k = 9), & \text{Ligurian surface longline} \end{cases}$$

- observation variance σ^2 of the catches

$$\sigma^2 \sim s(age, k = 3)$$

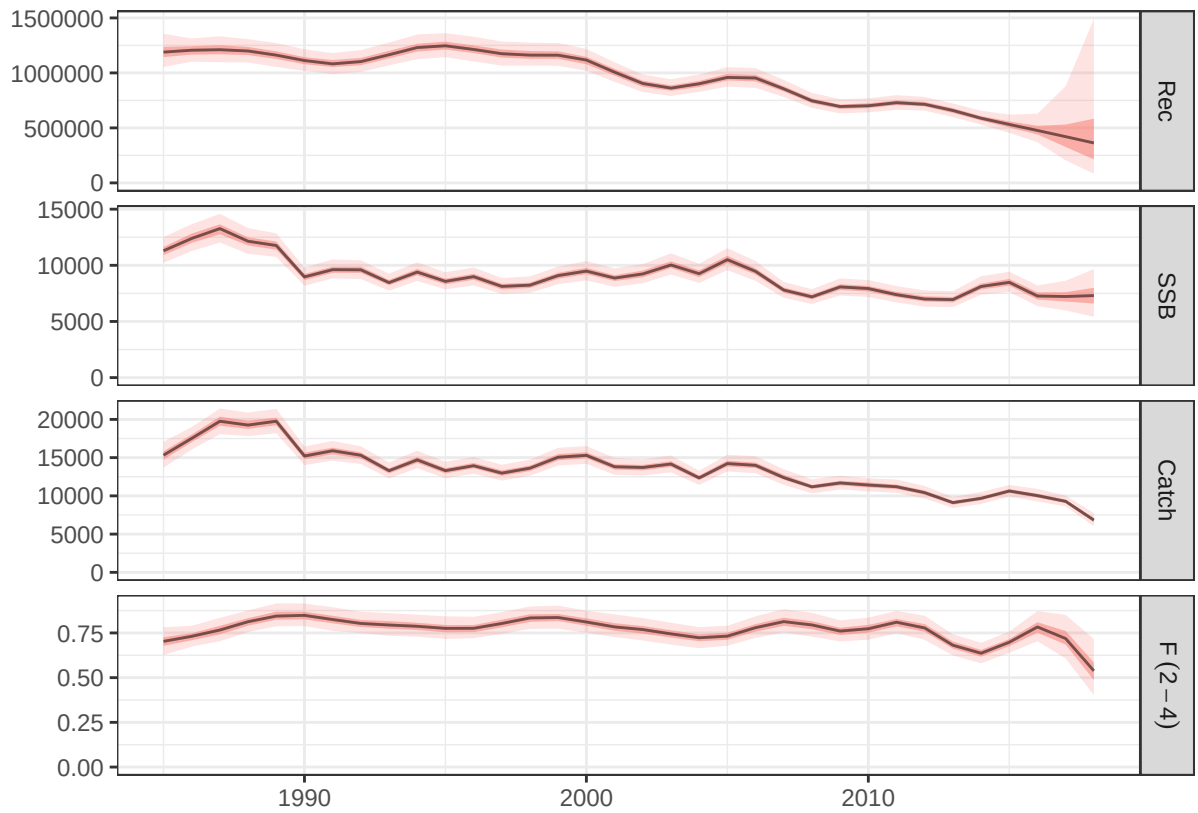
- variance of the indices

$$\tau_s^2 \sim 1, \quad \text{for all indices } s$$

- initial age structure $\mathbf{N}_1$

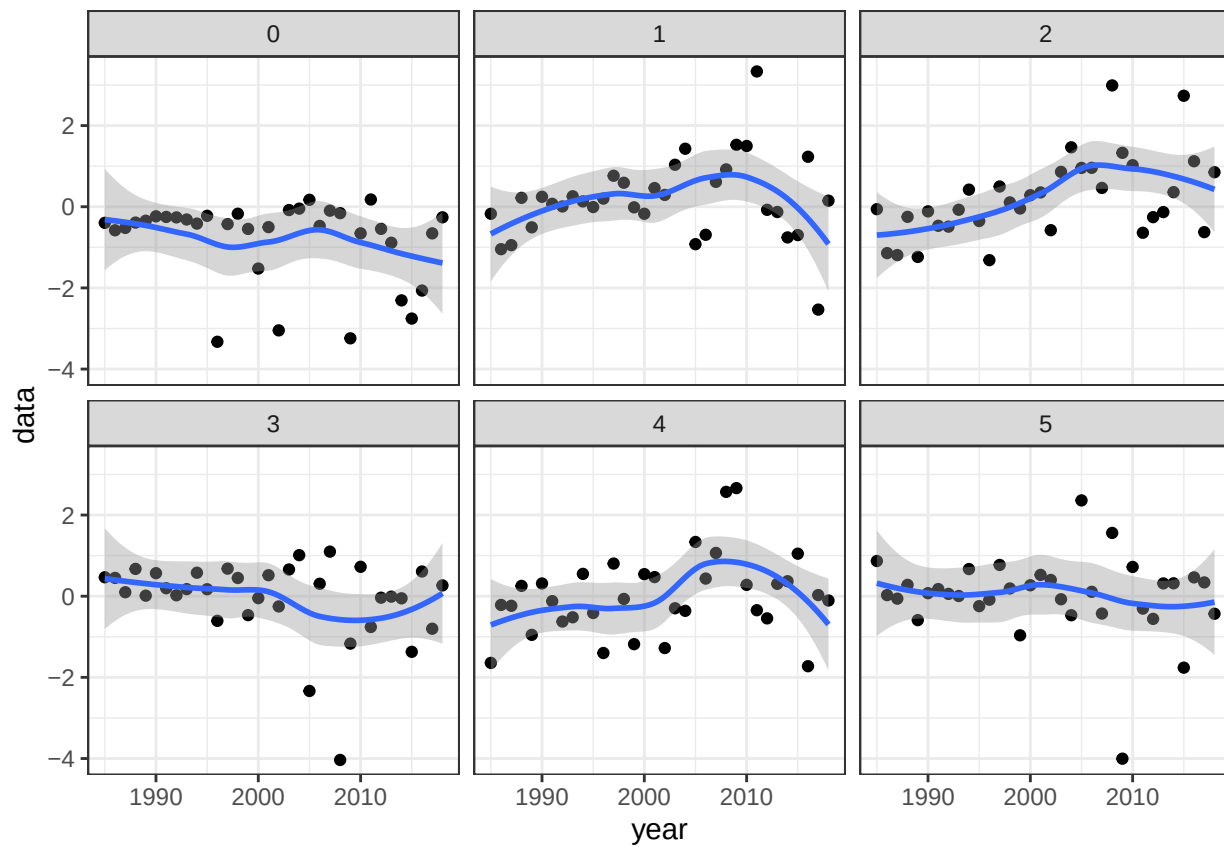
$$N : \sim s(age, k = 3)$$

final run | assuming constant M



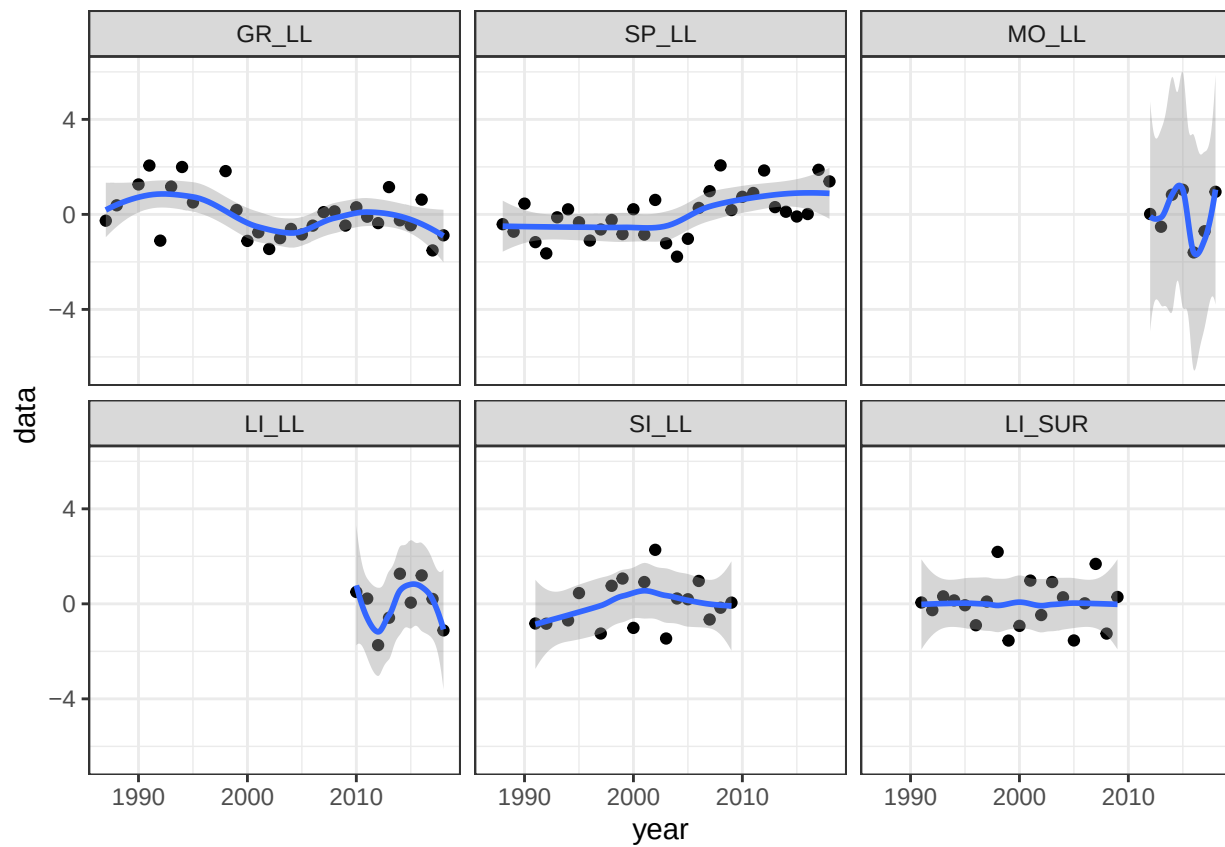
Diagnostics | final run assuming constant M

Standardized residuals of the catch numbers by age

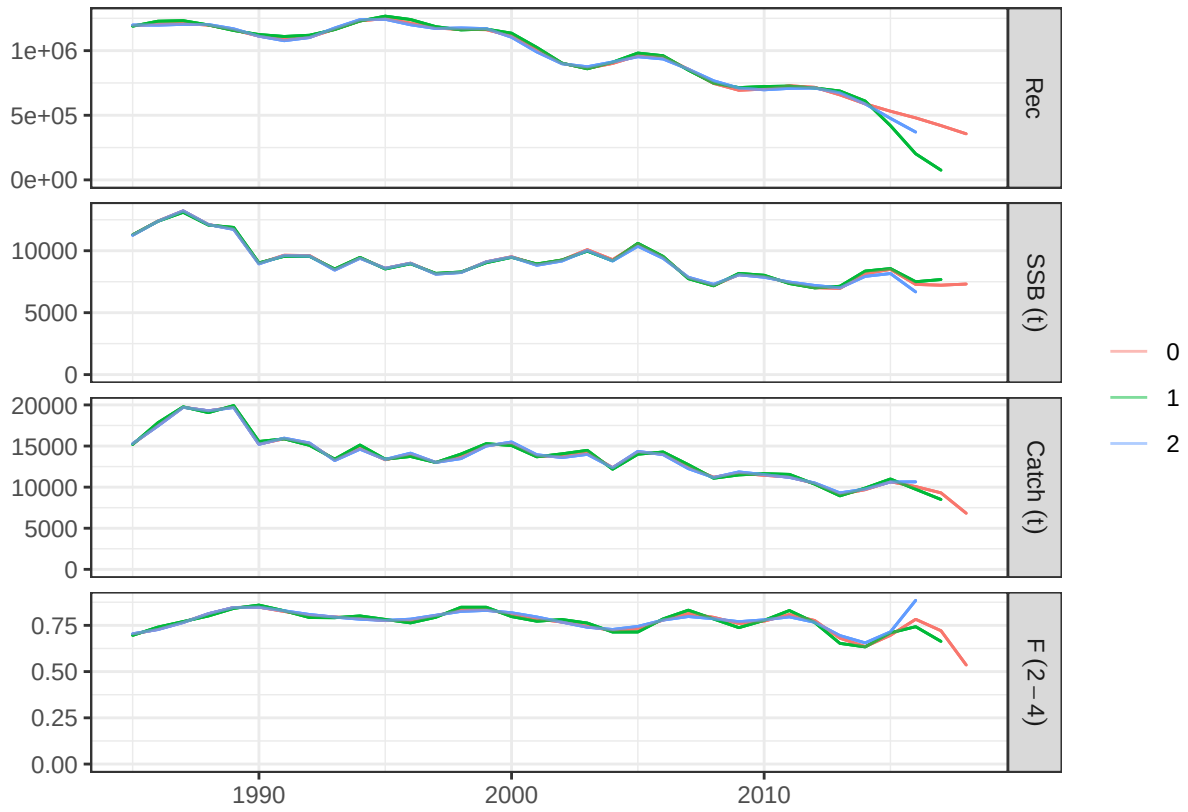


Diagnostics | final run assuming constant M

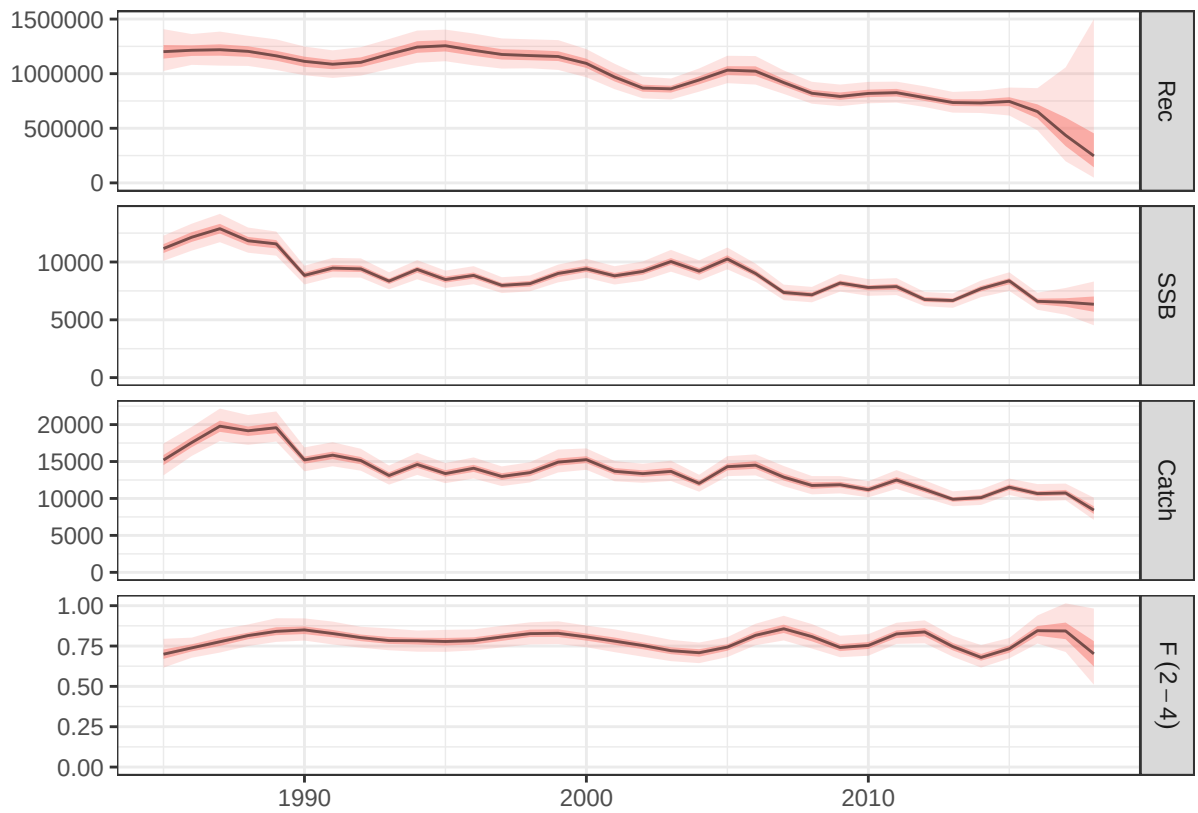
Standardized residuals of the CPUE series



Retrospective plot | final run assuming constant M

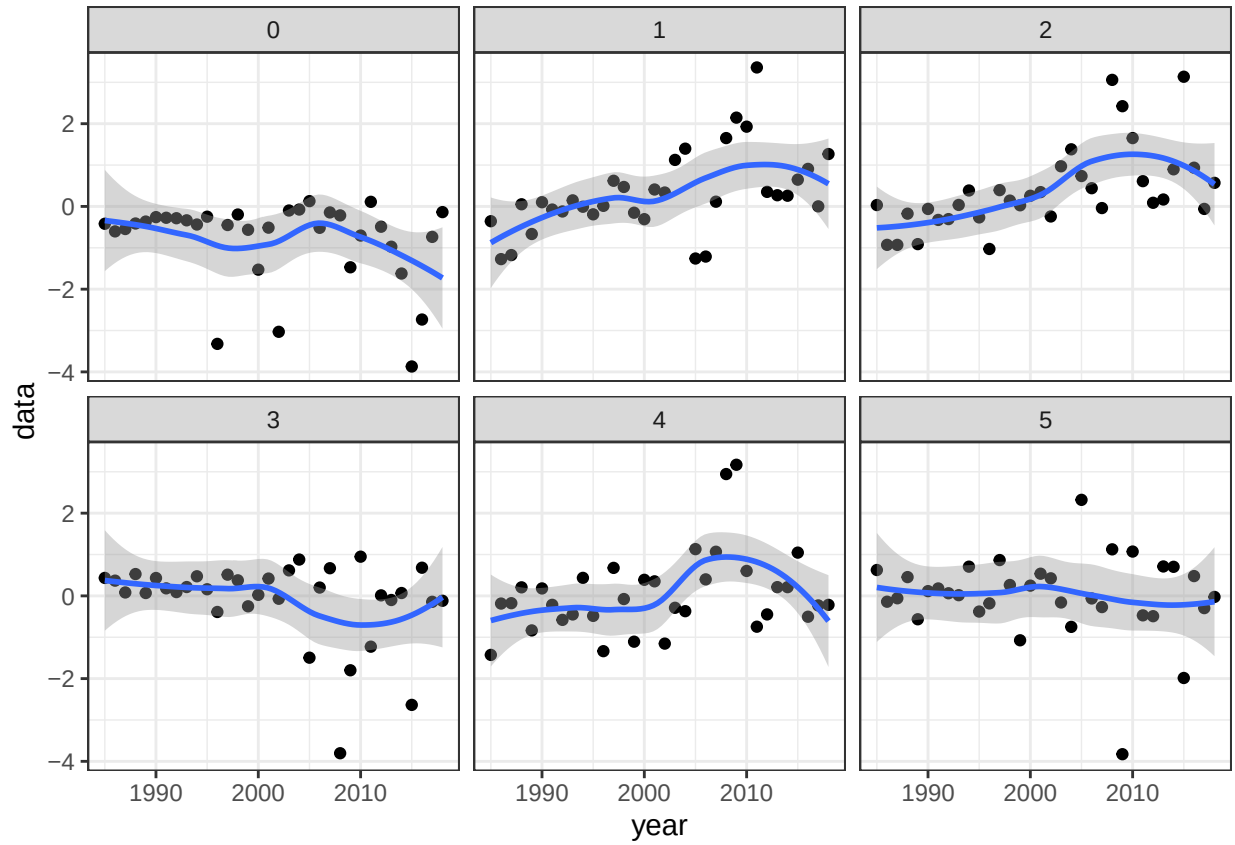


final run | assuming constant M discards included



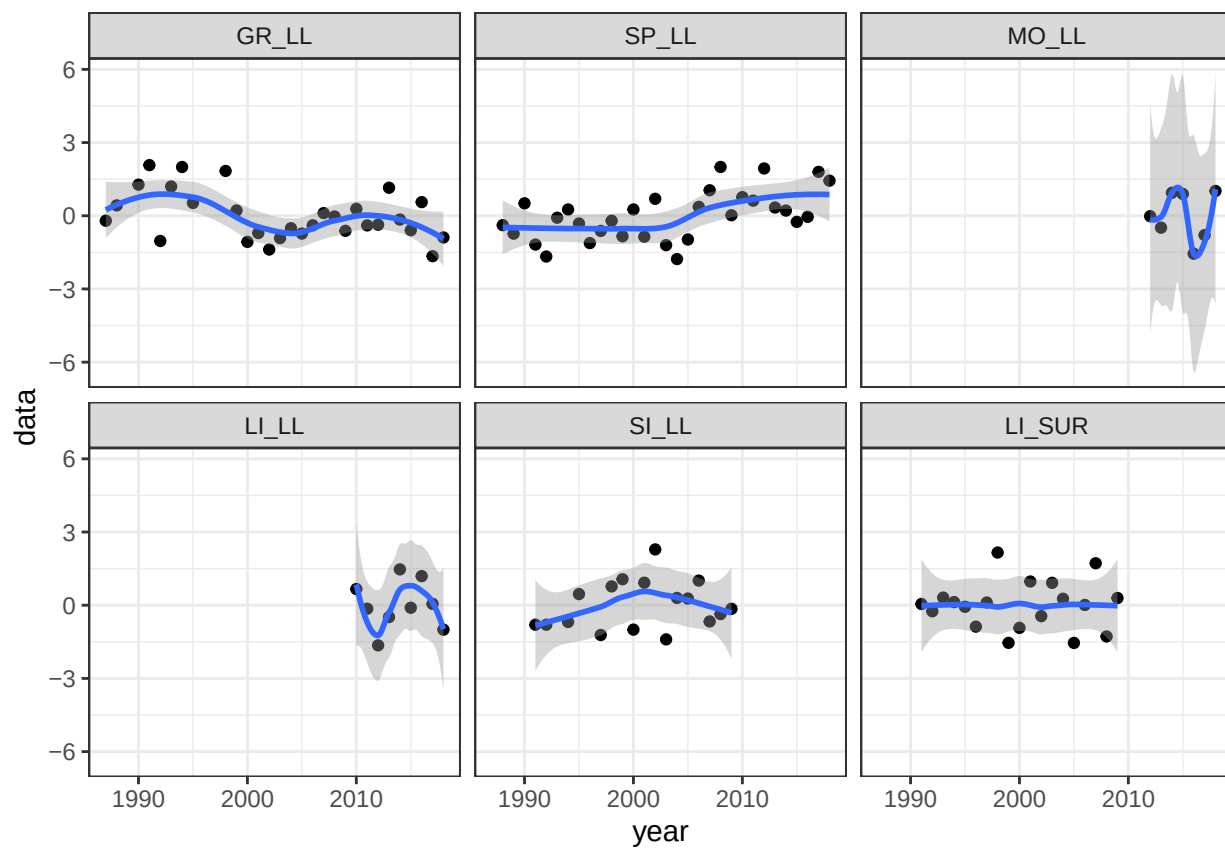
Diagnostics | final run assuming constant M discards included

Standardized residuals of the catch numbers by age

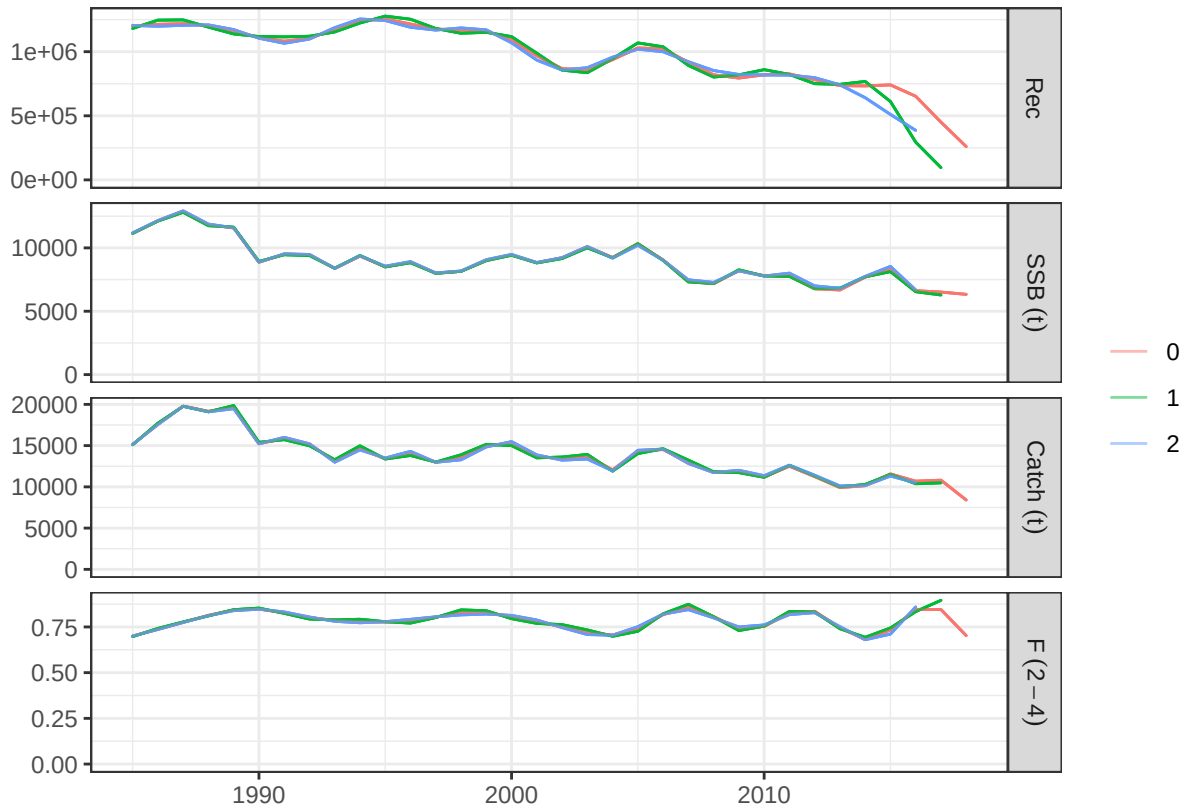


Diagnostics | final run assuming constant M discards included

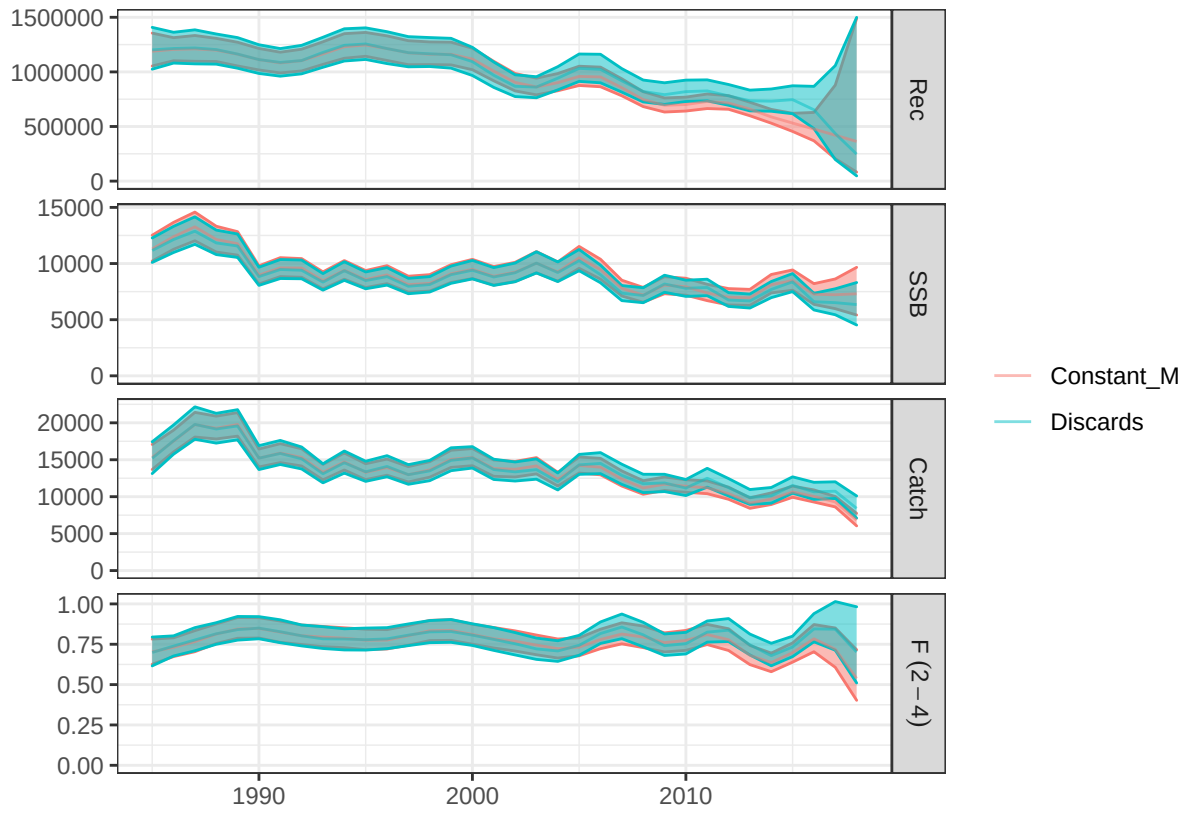
Standardized residuals of the CPUE series



Retrospective plot | final run assuming constant M discards included



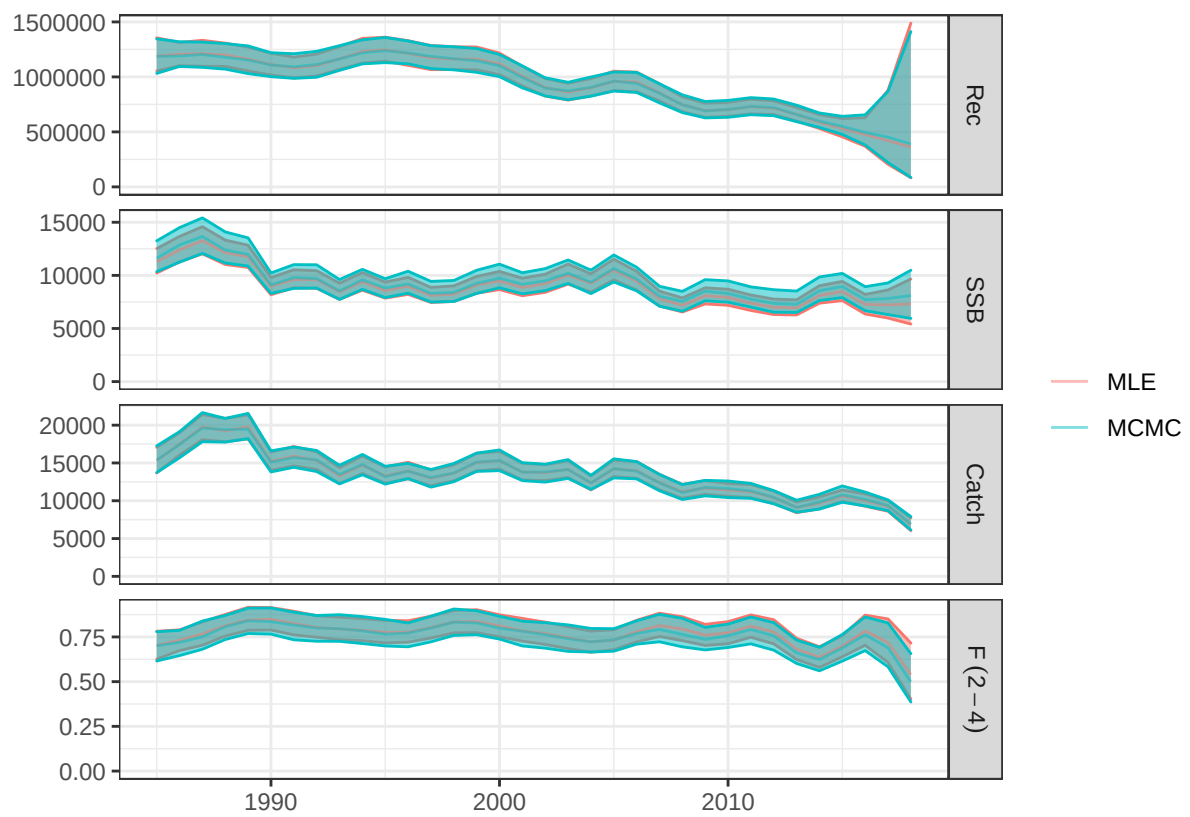
Comparison plot



Reference points

value	constantM	Disc_constM
Fmsy	3.833524e-01	3.753627e-01
Bmsy	4.586190e+04	4.351330e+04
F2018	5.355717e-01	7.024563e-01
SSB	7.312521e+03	6.331603e+03
FFmsy	1.397074e+00	1.871407e+00
FF0.1	2.618664e+00	3.535289e+00
BBmsy	5.001516e-01	5.177004e-01

MCMC | constant M



MCMC | constant M discards included

